Development of Nanomaterial Based Antimicrobial Coatings for Masks/Fabrics

The Centre for Materials for Electronics Technology is a premier research centre in Electronic materials under Ministry of Electronics and Information Technology, Government of India. C-MET is engaged in the development of process/product/technology and transfer of the technology in the area of Electronic materials, components and devices to cater to India's strategic and industrial applications, exploiting indigenous resources of raw materials.

The C-MET family comprises three Centres, located at Pune, Hyderabad and Thrissur with specialized research mandate at each place. C-MET, Pune is concentrating on the development of nanomaterials /composites, Lithium-ion batteries, LTCC packaging materials, materials for optoelectronic devices, materials for renewable energy. Its main areas of focus include development of nanostructured materials for various electronic applications.

The whole world is in search of the methods to stop the spreading covid-19. Now the only way to stop the spreading of covid-19 is the lockdown and keeping social distancing. Though the government agencies started vaccination but still it may require 2-3 years to complete the same due to lack of supply of vaccine and shortage of healthcare related manpower. Further the vaccinated person may also act as asymptomatic carrier and can be a super spreader of covid-19. The CORONA virus remains active on the surfaces, in the air from couple of hrs to few days. Majority of the people are getting infected by the virus by touching or getting in contact with the different kinds of surfaces unknowingly. The front-line workers, hospital staff have the major risk as they are providing treatments to the COVID-19 infect persons in the hospitals. The developed mask and cloths are not healing the corona virus if it comes in the virus contacts, but it is only interrupted the path to enter the virus in the human's body. Means the present preventive mask and cloths are used as obstacle to the virus and not the destroyer.

In this regard C-MET,Pune have developed the self-healing mask and cloths by thin coating of the nanoparticles. These mask and cloths not only interrupt the path of viruses to enter in the person's body but it will destroy the viruses too. The nanoparticles have antiviral, antibacterial and antifungal properties. The antimicrobial properties of developed nanomaterials coated mask were tested against bacillus and S. Aureus bacteria and found effective for its growth inhibition.

About the product:

C-MET, Pune has developed nanomaterial-based coatings for the mask and other healthcare fabrics. The mask having size of 4x8 inch requires around 40 to 45 mg of nanomaterial coatings. The process for nanomaterial synthesis and its coatings have been optimized for 10 g batch scale and can be scale up easily. In 10 gm nanomaterials almost 200-250 mask can be coated. The coatings remained on the masks/fabrics even after several wash. The nanomaterials is in the powder and need to be mix with polymer solution for performing the coating.

This technology is specific for making the antimicrobial coated mask and other fabrics useful in health care industries.

Technology Transfer

C-MET, Pune would like to invite competent industrial partners for commercialization of the above product (Nanomaterial coatings) through transfer of technology. The technology will be transferred on non-exclusive basis. The ToT package contains the following

1. Training related to material synthesis
2. Training related to the coating of mask

The level of technology readiness at C-MET

The technology readiness level is 6

Period of training under the agreement

C-MET, Pune will give one week training to the staff/engineer of the ToT partner at the premises of C-MET, Pune.

Who can apply?

Any industries who planned to buy this technology can approach C-MET, Pune for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology. A Committee of experts constituted by the C-MET will assess capabilities and strengths of the industry before finalizing the technology partners.

How to apply?

Interested companies may send expression of interest with their details by filing the following questionnaire along with supporting documents to:

Programme co-ordinator
Centre for Materials for Electronics Technology,
Off Pashan road, Panchawati, Pune-411008, India
Email:prog.coord@cmet.gov.in

General:

The industry willing to take technology for commercial exploitation will be required to enter into a ToT agreement with C-MET, Pune as per the terms and conditions approved by the competent authority in the MeitY prescribed format.